



Nuno de Sousa

Data Scientist & Subject Matter Expert

Quantitative Strategy, Pricing, Forecasting & AI

PhD | Executive MBA Candidate

Based in Oporto, Portugal

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Portuguese | Male

Born 28 July 1982

nunodsousa.github.io/cv | [LinkedIn](#) | [GitHub](#) | [ORCID](#)

English CV

EXECUTIVE PROFILE

Data Science and AI leader with over 10 years of experience combining **quantitative expertise** with **business strategy and organizational leadership**. Currently works as a **Data Scientist and Subject Matter Expert**, developing **pricing and forecasting solutions** for complex, data-sparse markets. Has led **multidisciplinary teams of more than 10 professionals** and delivered **strategic analytics and AI initiatives** for international clients across energy, finance, chemicals and industrial sectors. Holds a **PhD in Theoretical Physics** and is completing an **Executive MBA**, bringing together analytical rigour, executive communication and a growing focus on **strategy, management and business leadership**.

LEADERSHIP AT A GLANCE

Strategy & Delivery

Data and AI strategy, solution architecture, prioritisation, business-technology alignment and delivery of analytics initiatives.

People Leadership

Leadership of multidisciplinary teams of more than 10 professionals, cross-functional coordination and technical direction.

Stakeholders & Growth

Executive communication, strategic advisory, pre-sales and client-facing solution design.

PROFESSIONAL EXPERIENCE

Data Scientist | Subject Matter Expert in Quantitative Modelling, Pricing & Forecasting [AuctionConnect](#)

Mar 2026–Present

Oporto / Remote

- Defines the analytical approach for nowcasting and forecasting marine bunker fuel prices across global ports, aligning model development with commercial priorities and market needs.
- Transforms fragmented market data, economic indicators and pricing signals into actionable intelligence for commercial planning, risk assessment and strategic decision-making.
- Develops robust statistical and machine learning frameworks for complex, data-sparse markets, balancing analytical accuracy with business applicability.
- Communicates model assumptions, outputs and market implications to business stakeholders and contributes specialist knowledge to the long-term evolution of pricing products.

Data & AI Lead | Subject Matter Expert [DataJuicers](#) and [Simia-Tech](#)

Nov 2019–Mar 2026

Madrid / Oporto / Remote

- Led cross-functional teams of more than 10 data scientists, engineers and analysts, aligning business objectives, analytical rigour and timely delivery for international clients.
- Architected production-grade Data Science and AI platforms and advanced models for pricing optimisation, demand forecasting, inventory planning and market intelligence.
- Served as the business-technology interface and strategic advisor to senior stakeholders, translating analytical outputs into clear decision support.
- Supported pre-sales through solution design, technical validation and client-facing presentations, helping secure and expand long-term engagements.

Quantitative Analyst [Arfima Trading](#)

Apr 2019–Oct 2019

Madrid

Designed and deployed automated futures trading strategies from end-of-session to high-frequency horizons. Built statistical and machine learning models, a custom backtesting framework and integrated risk management protocols.

Data Scientist [DataJuicers](#)

Jun 2017–Mar 2019

Madrid

Delivered big-data, time-series forecasting and machine learning solutions across luxury, retail, services and industry. Led a TensorFlow computer-vision recommendation project recognised by Google and contributed quantitative analysis to hedge-fund projects.

Researcher / Lecturer [Universidad Autónoma de Madrid](#) & [Donostia International Physics Center](#)

2009–Jun 2017

Spain

Taught Physics and Informatics Engineering, supervised research students, and conducted research in Computational Physics, Electrodynamics and Mathematical Methods.

EDUCATION

Executive MBA

Oct 2024–Expected Oct 2026

- Católica Porto Business School - CPBS
- Strategic leadership and decision-making: executive decisions, competitive strategy and business model innovation.
 - Corporate finance and risk management: financial analysis, valuation, capital allocation and enterprise risk.
 - Operations and organisational performance: operational excellence, performance management and large-scale transformation.
 - Executive communication and governance: negotiation, stakeholder management and corporate governance frameworks.

PhD in Theoretical Physics

Sep 2010–Oct 2014

Universidad Autónoma de Madrid

Condensed Matter Physics and Nanotechnology. Dissertation: *Light scattering in disordered and nonreciprocal media*. Awarded *Sobresaliente Cum Laude*, the maximum classification.

Master in Photonics

Sep 2009–Jun 2010

Universidad Autónoma de Madrid

GPA 8.8/10. Thesis on light-emission statistics in correlated random photonic nanostructures.

Licentiate Degree in Physics

Universidade do Porto

LEADERSHIP, BUSINESS & TECHNICAL CAPABILITIES

- Leadership & Management
- Strategy & Delivery
- Data and AI strategy, solution architecture, initiative prioritisation, delivery governance and business-technology alignment.
- People Leadership
- Multidisciplinary team leadership, cross-functional coordination, technical direction, mentoring and quality ownership.
- Stakeholders & Growth
- Executive stakeholder communication, strategic advisory, pre-sales, negotiation and client-facing solution design.
- Business Applications
- Pricing optimisation, demand forecasting, inventory planning, market intelligence and quantitative trading.

- Quantitative Modelling
- Pricing models, time-series forecasting (ARIMA, ETS, state space and Prophet), Bayesian modelling, simulation, optimisation, survival analysis and customer lifetime value.
- Machine Learning & AI
- scikit-learn, XGBoost, TensorFlow, LLM applications, RAG, model evaluation and agentic workflows.
- Data & Production Systems
- Python, SQL, pandas, NumPy, statsmodels, Docker, Git, Azure and workflow orchestration with n8n.

RESEARCH, TEACHING & RECOGNITION

21

peer-reviewed publications

~50

international oral presentations

5

theses supervised

8

science research projects

- Director of the *AI for Marketing* programme at Católica Porto Business School.
- Led the 2017 Dress Recommendation System, recognised by Google for its application of TensorFlow and computer vision.
- Supervised three master's theses and two bachelor's dissertations across Physics and Informatics Engineering.
- Research spans nanophotonics, magneto-optics, computational physics and light-matter interactions; the full record is available online.

PROFESSIONAL DEVELOPMENT & LANGUAGES

- Selected certifications
- Python for Financial Analysis and Algorithmic Trading - Udemy, 2018
- Machine Learning Specialization - University of Washington / Coursera, 2016–2017
- Applied Plotting, Charting and Data Representation in Python - University of Michigan / Coursera, 2017
- Inferential Statistics - Duke University / Coursera, 2017
- Languages
- Portuguese - Native
- English - Certified C1+ (Aug 2024)
- Spanish - 12 years living in Spain